Face & Profile Milling

Recall the different movements on a milling machine and the following from I2E:

Up Cut/Down Cut

Cutting Equations

$$V_c = \frac{\pi D N}{1000}$$

$$F = N \times f_n \times z_n$$

- Cutting Speed & Spindle Speed
  - Cutting Speed: Surface speed of cutting edge as it enters workpiece; tool oriented; dependent on the tool's metrics
  - Spindle Speed: rotation speed of the motor; machine oriented; determined by machine based on necessary cutting speed.

Face Milling Methods

- One pass:
- Zig Zag:
- One Way:
- Dynamic:
- During face, try to ensure full contact during each pass instead of parallel passes, (e.g. by ensuring each pass covers the full surface, using dynamic facing, or one pass facing)
- Avoid facing over holes and slots as they are quite demanding; instead do drilling at the end.
- If inevitable, reduce feedrate by 50% during facing.
- Radial passes should be used to keep the cutter moving, avoiding dwell and chatter.

Profile Milling

- Roughing involves multiple passes, finishing involves one singular pass

During profile milling, we adjust the approach and retract/lead in and lead out/engage and disengage.

there are 4 main methods:

without approach and retract: uses plunging, requires central cutting edge, cutter mark left at entry/exit point

approach and retract with arc modes: usually for closed contours, with 90 deg arc, smooth transition provides good surface finish

approach and retract with arc and line modes: same as arc modes, but cutter starts and ends at same place. Pre drilled hole may be done for initial plunging

Tangent Straight lines for open profiles: follows profile tangent, good surface finish

Cutter Compensation

cutter has some diameter; needs to be offset from the contour path to prevent undercutting.

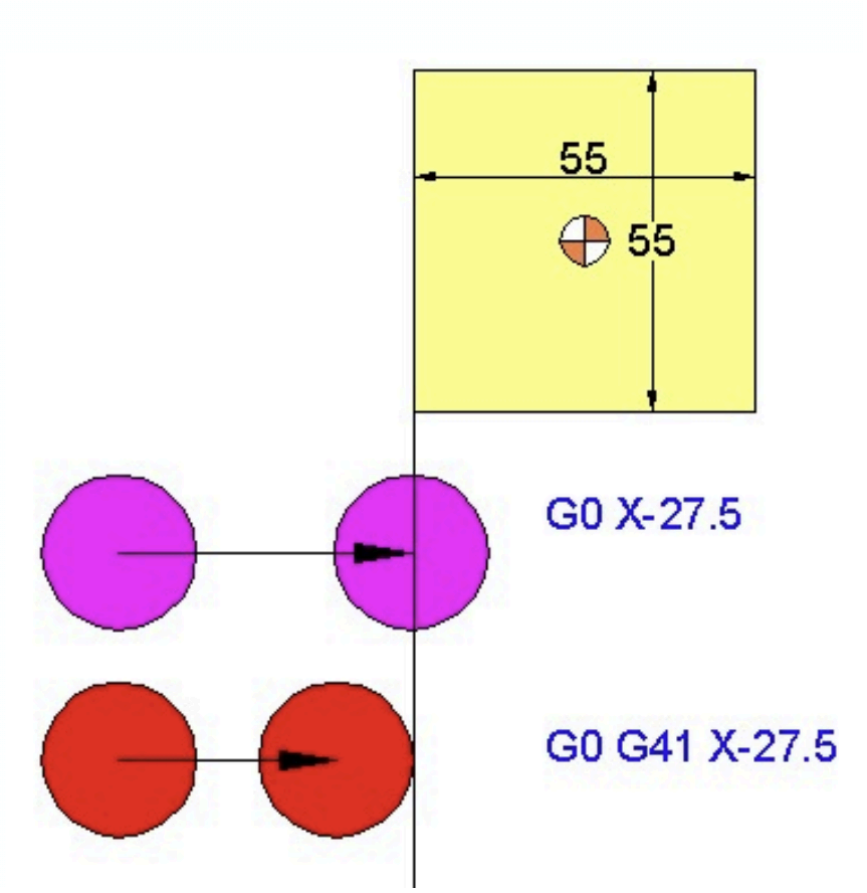
e.g. if the cutter has diameter 25 and it is cutting a 55mm square profile, it follows a 55+25=80mm square path

(typically the actual cutter diameter is less than the nominal value, thus we use G40-G42 for cutter compensation

G41 is when the cutter is moving on the left of the contour, while G42 is when the cutter is moving on the right of the contour.

G40 removes tool compensation and can be used for slot milling or engraving

G41/G42 is used in the G code to automatically offset the tool



There are also computer/control compensation systems:

computer: calculated by CAM software, fixed and runs one to one from CAM software to machine; poorer dimensional accuracy thus better for roughing

control: tool path is generated by CAM controller with G41/42; allows actual tool input for more accurate dimensions; used for finishing